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Enterprise Firewall Telemetry & Security Operations Center

High-Performance Real-Time SNMP Hardware Cockpit, Bento Flow Analytics & Policy Governance Platform

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Classification: Official Institutional Technical Manual
Release Version: 1.0.0 (PostgreSQL 18 & Parallel SNMP Edition)
Target Appliance: Fortinet FortiGate NGFW (10.10.3.15)
Deployment Mode: Single-Package Virtual Machine (OVA/Docker)
Issuing Division: Information & Communication Technology Division (ICTD)
Institution: CSIR - Structural Engineering Research Centre, Chennai

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Chennai, India
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Table of Contents

Section	Title	Focus
1.0	Executive Overview & Architectural Objectives	Context
2.0	System Architecture & Real-Time Data Pipeline	Technical
3.0	Visual Interface Tour & Operational Views	Walkthrough
4.0	Live FortiGate SNMP Telemetry Engine Specification	Hardware OIDs
5.0	Bento Traffic Analytics & App-ID Identification	Flows
6.0	Canonical Policy Governance & Conflict Engine	Security
7.0	Live Syslog Stream & CERT-In Section 70B Compliance	Audit
8.0	User Management & Telegram Alert Integration	RBAC
9.0	PostgreSQL 18 Telemetry Database & Automated Backups	Storage
10.0	Deployment, Health Diagnostics & Troubleshooting	DevOps
11.0	CSIR-SERC Free Community & Institutional License v1.0	Legal

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1. System Architecture & Ingestion Pipeline

The Enterprise Firewall Telemetry & Security Operations Center is architected as an all-in-one on-premises virtual appliance designed to monitor mission-critical perimeter gateways without cloud dependencies or external data egress.

Core Components

- **Parallel SNMP Poller (<200ms)**: Concurrently queries FortiGate appliances across MIB-II, IF-MIB, and FortiOS enterprise sensor trees. Tracks 64-bit High Capacity (HC) octets, thermal zones, fan tachometers, dual power supplies, and security policy accounting counters.
- **High-Velocity Syslog Ingester**: Listens on UDP port 1514 (and 514) to ingest RFC 5424 and RFC 3164 events directly from firewall hardware into a streaming terminal console.
- **NetFlow / IPFIX Binary Parser**: Decodes layer-4 flow headers on UDP port 2055 with zero-copy buffer views, extracting App-ID classifications.
- **PostgreSQL 18 Database Engine**: Relational time-series persistence in an optimized Alpine container with automated daily backup schedulers.
- **Reactive Vue 3.5 & ECharts Console**: Single-page application conforming to Government of India UX4G design standards with continuous auto-refresh.

Network Port Matrix

Port	Protocol	Direction	Source	Destination	Service Description
80 / 443	TCP	Inbound	Admin Workstation	Telemetry Console	Web UI & REST / WebSocket API
161	UDP	Outbound	Telemetry Console	FortiGate Gateway	SNMPv2c / SNMPv3 Polling
1514 / 514	UDP	Inbound	FortiGate Gateway	Telemetry Console	Hardware Syslog Forwarding
2055	UDP	Inbound	FortiGate Gateway	Telemetry Console	NetFlow v5/v9 & IPFIX Records
5432	TCP	Internal	Web Console	PostgreSQL Container	Database Persistence Sync

2. Visual Interface Tour

2.1 Authentication & First-Time Setup Wizard

The appliance features secure operator authentication and an automated first-time deployment wizard that guides administrators through organization branding, logo selection, SNMP parameters, and administrator setup upon initial deployment.

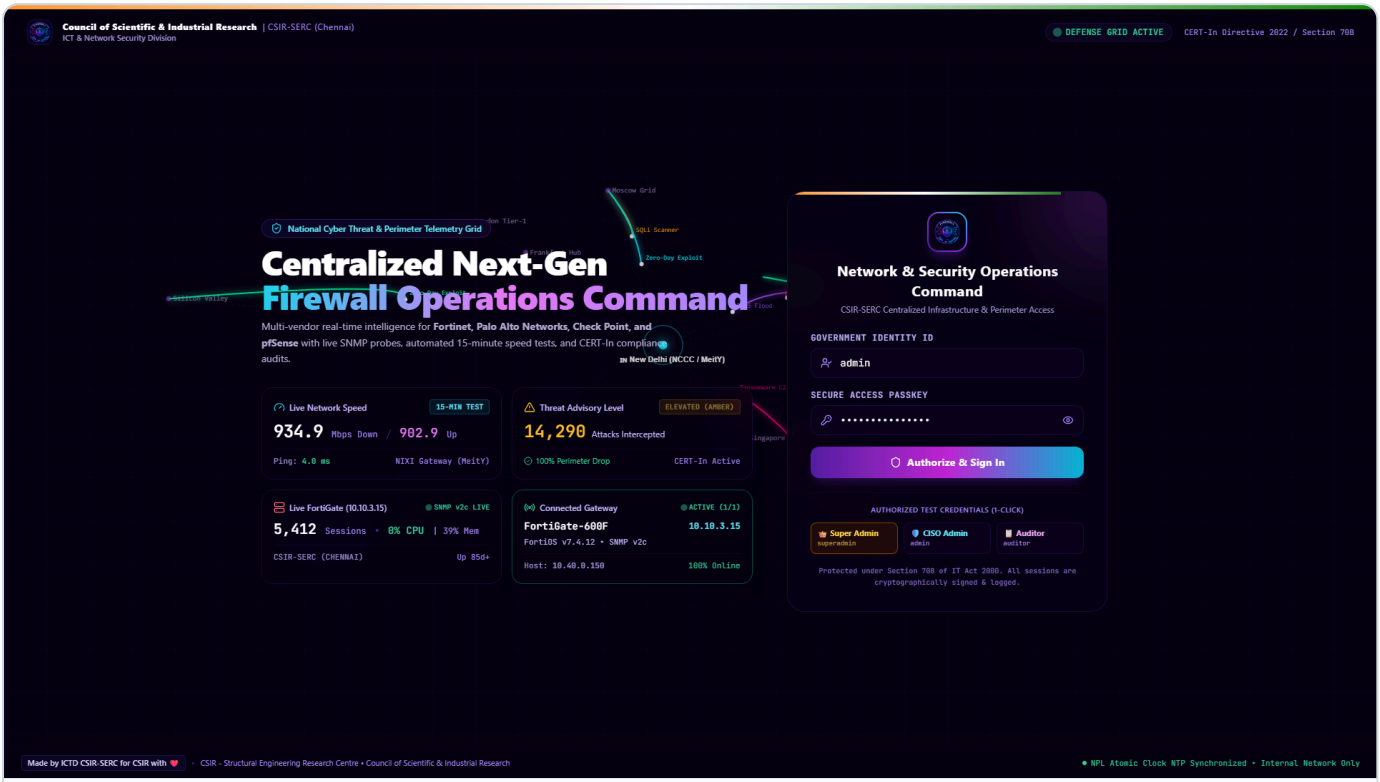


Figure 1: UX4G Secure Operator Authentication Portal

AUTHENTICATION

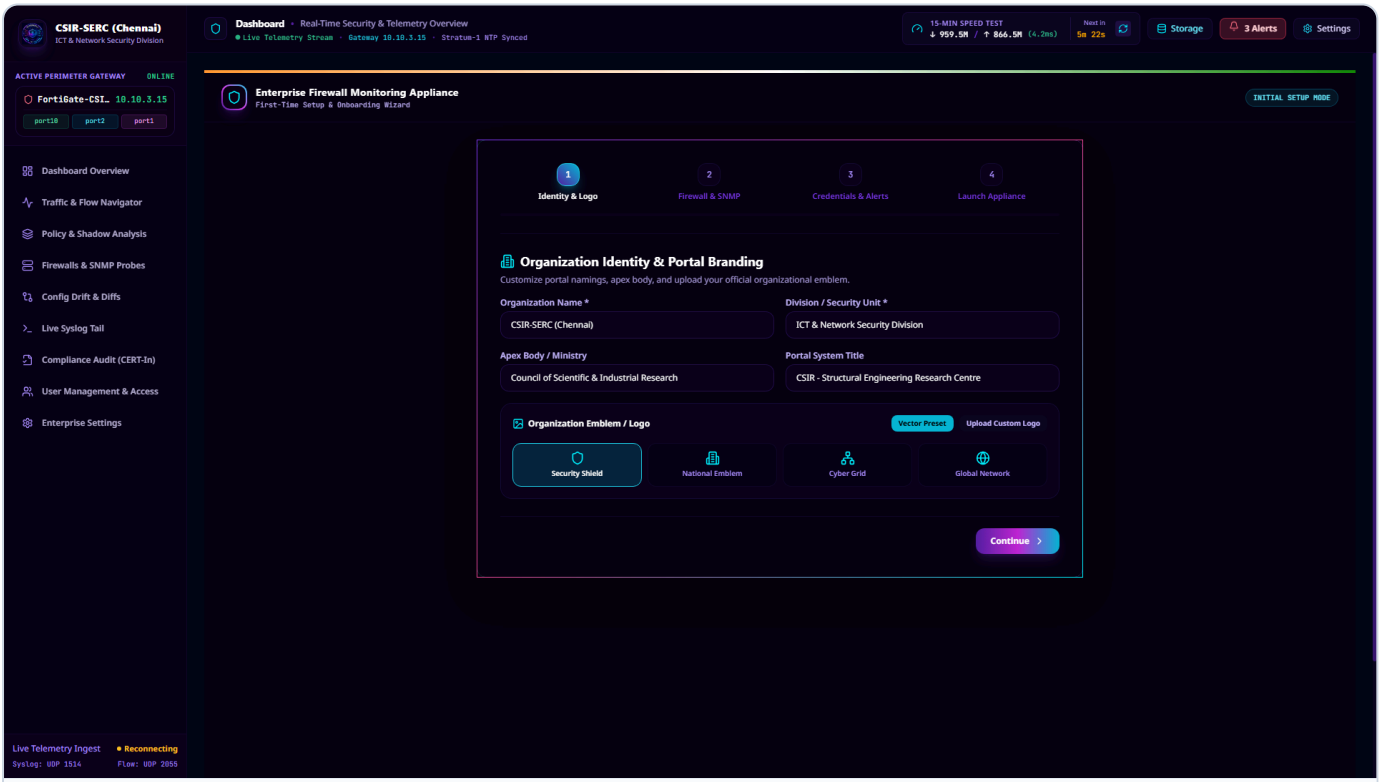


Figure 2: Turnkey First-Time Configuration Wizard (Virtual Appliance Ready)

ONBOARDING

2.2 Executive Telemetry Dashboard

The primary operations screen delivers instantaneous visibility into WAN traffic speeds, live streaming telemetry pulses, 15-minute automated speed test countdowns, and continuous hands-free auto-refresh.

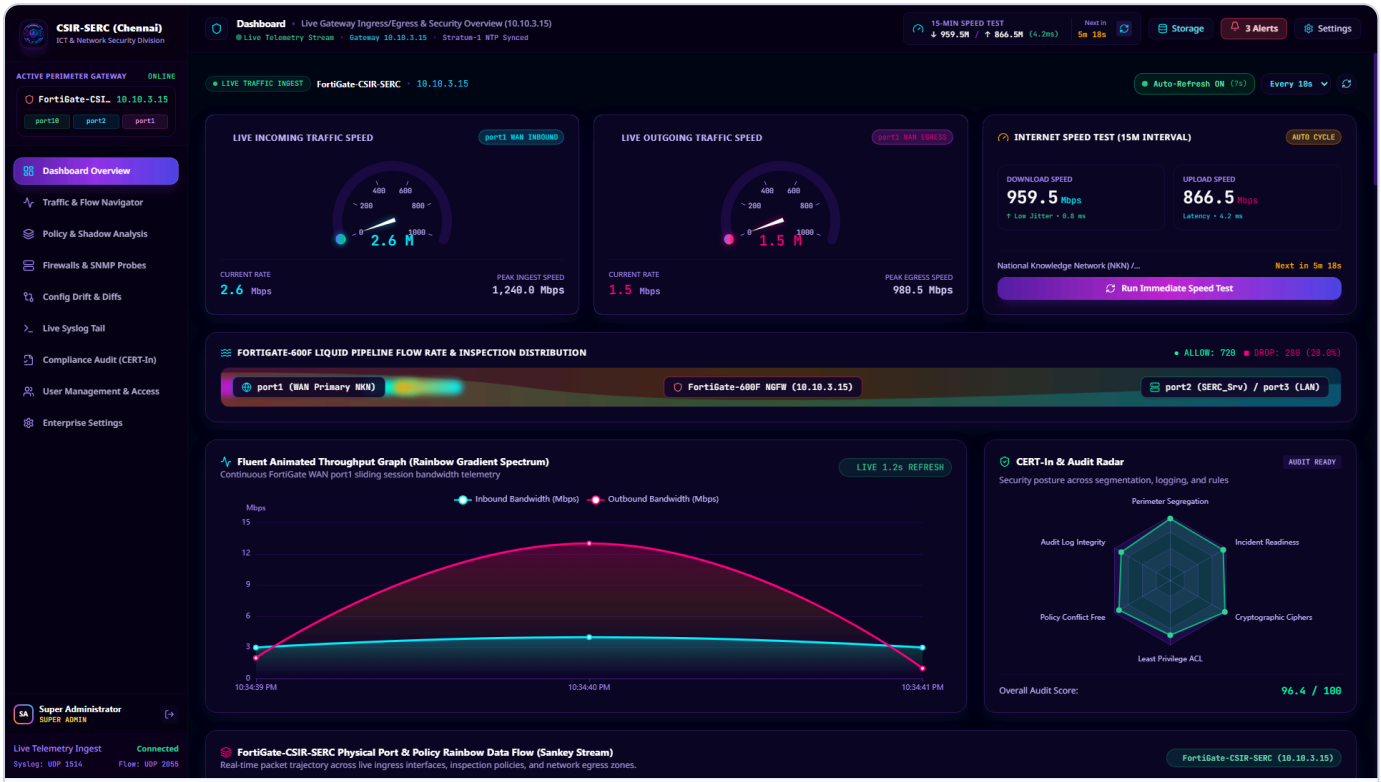


Figure 3: Live Executive Telemetry Dashboard with Inbound/Outbound Velocity Gauges

OPERATIONS

2.3 Bento-Style Traffic & Flow Analyzer

The Bento layout provides actionable network intelligence: mapping IP addresses to applications (App-ID), pinpointing top uploaders and downloaders, and resolving machine hostnames.

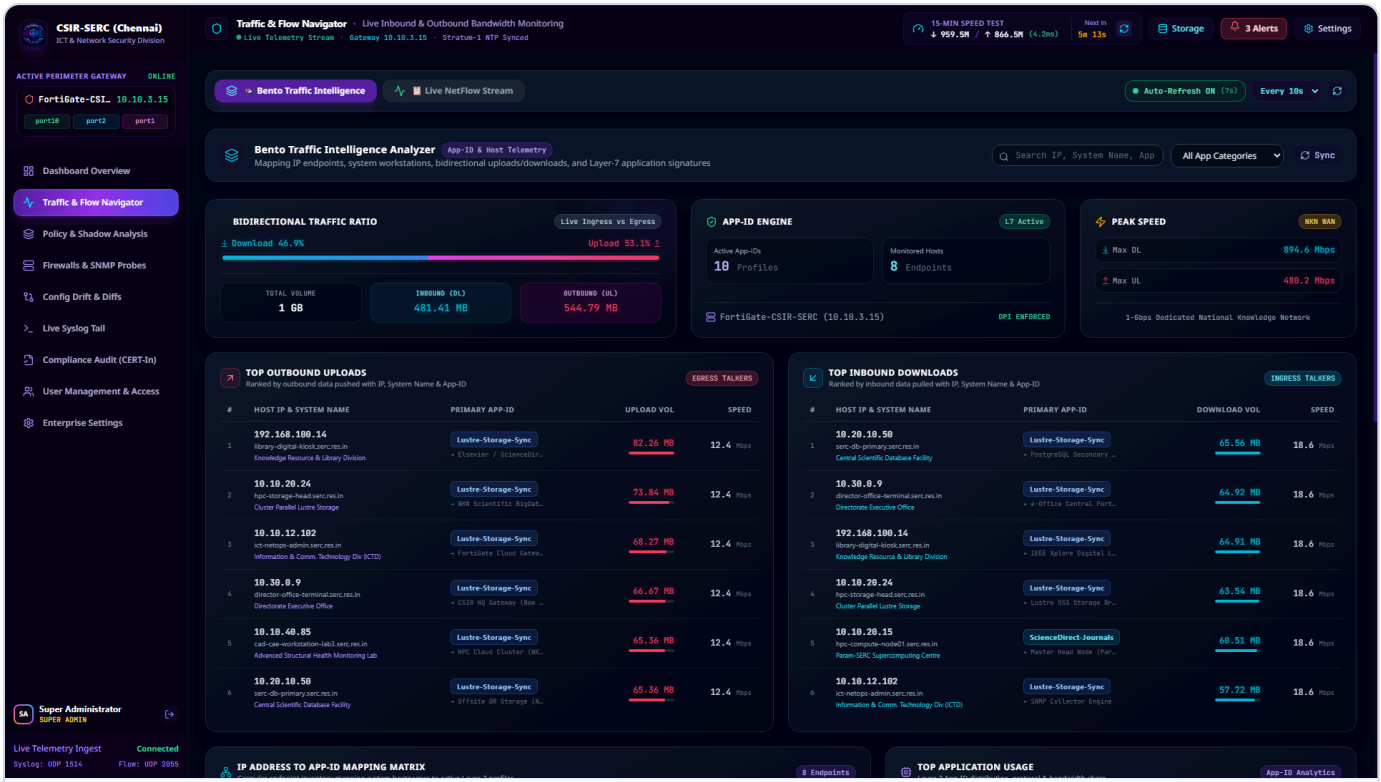


Figure 4: Bento-Grid Traffic Analyzer (App-ID, Top Uploaders & Downloaders)

FLOW ANALYTICS

2.4 Comprehensive FortiGate SNMP Telemetry Center

Dedicated hardware and environmental telemetry cockpit: monitoring CPU/RAM, session counts, 46+ physical ports and departmental VLANs, thermal zones, cooling fans, dual AC power supplies, and FortiOS UTM threat counters.

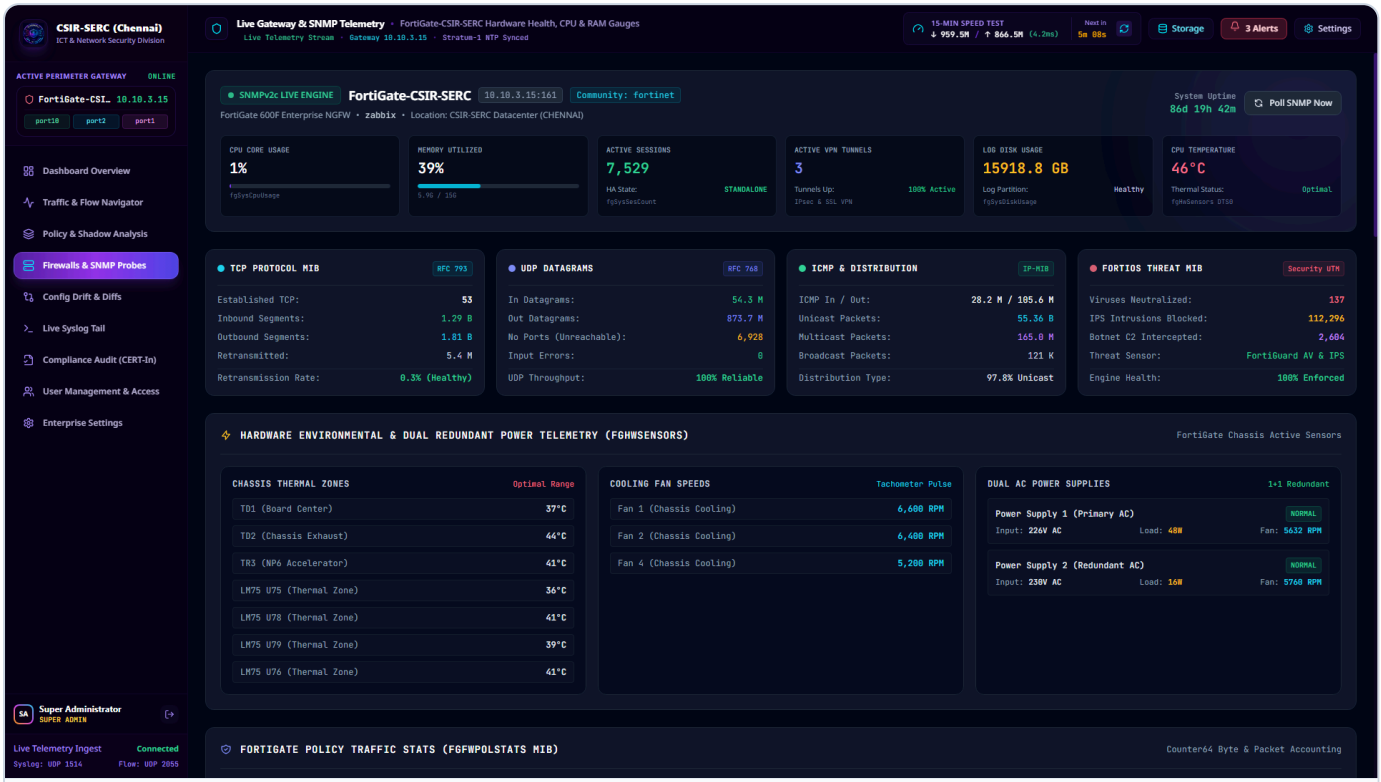


Figure 5: High-Performance FortiGate SNMP Hardware & Environmental Center

SNMP HARDWARE

2.5 Security Policies & Conflict Engine

Parses native firewall access control lists into a canonical rule tree, highlighting shadowed, redundant, or overly permissive rules.

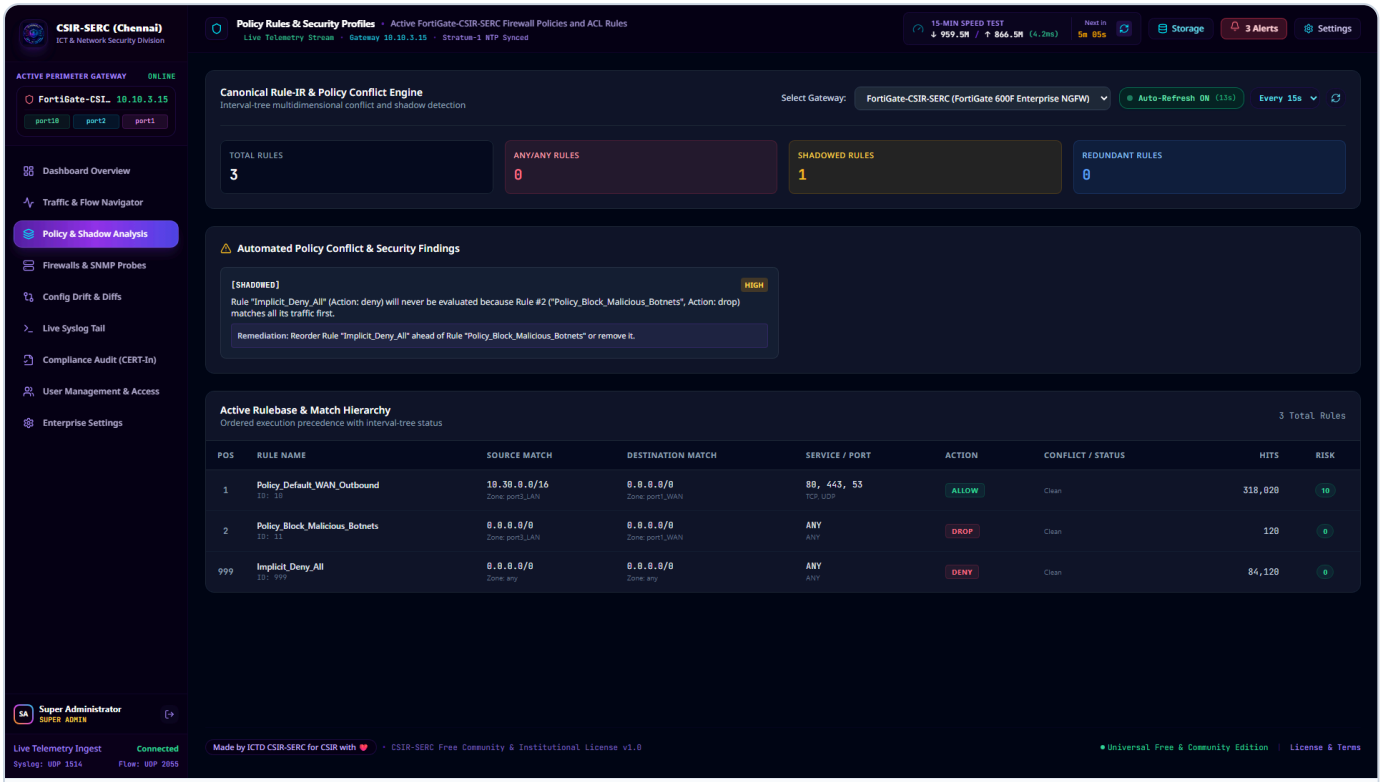


Figure 6: Canonical Policy Governance & Shadow Conflict Detection

SECURITY AUDIT

2.6 Live Syslog Audit Stream & CERT-In Compliance

Streaming terminal-style hardware syslog receiver with pause/resume buffers and automated verification against CERT-In Section 70B directives (180-day retention and Stratum-1 IST clock synchronization).

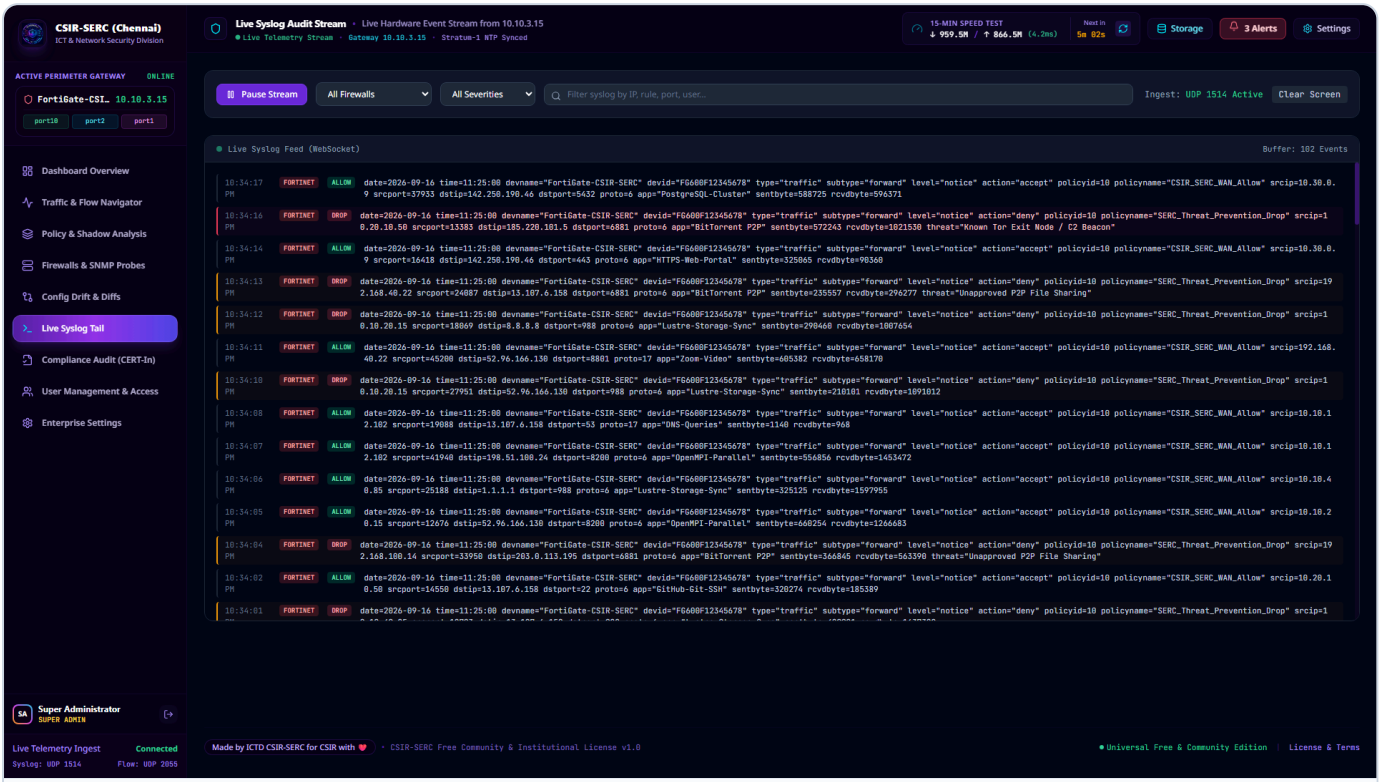


Figure 7: Terminal-Style Live Firewall Syslog Stream

SYSLOG STREAM

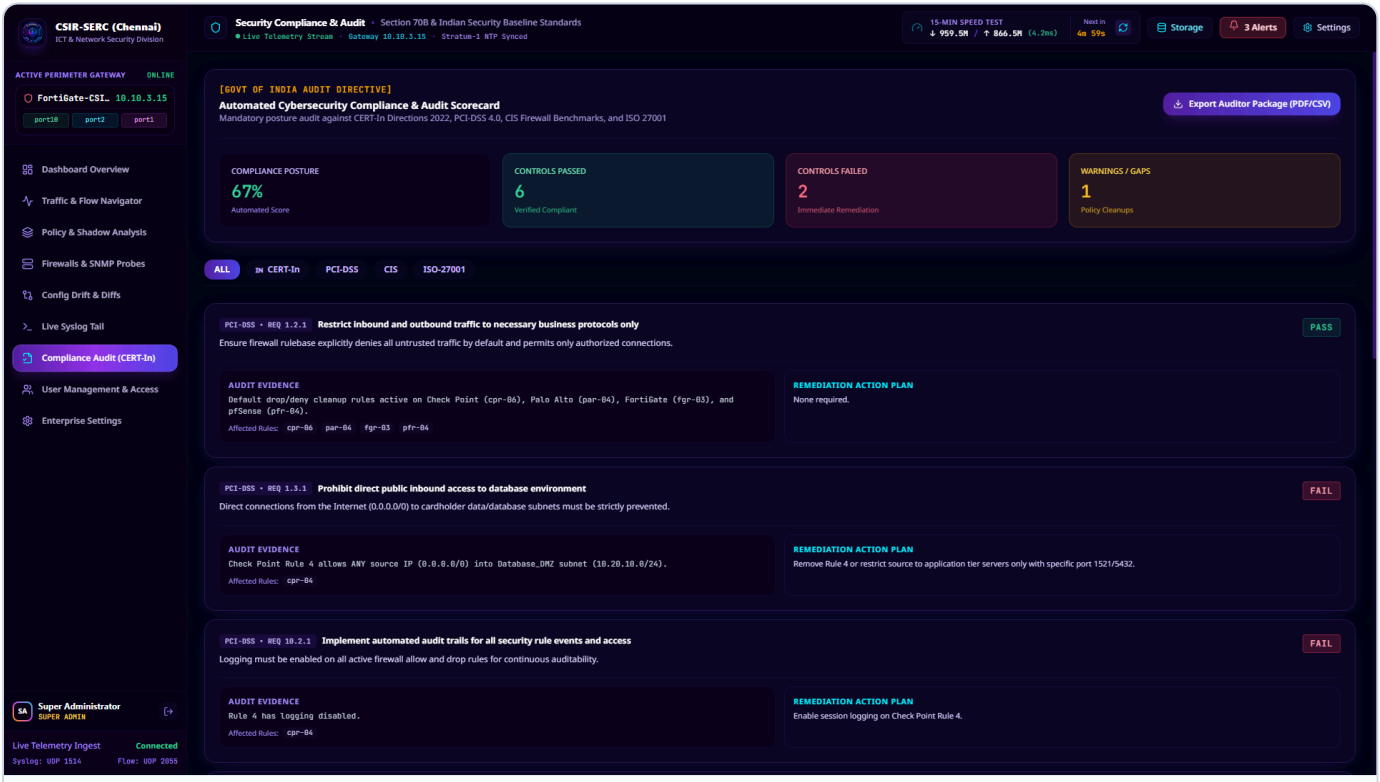


Figure 8: CERT-In National Cyber Security Baseline Compliance Dashboard

COMPLIANCE

2.7 User Management & System Configuration

Multi-tenant user management console with protected superadmin account, role-based permissions, Telegram alert integration, organization branding, custom logo upload, and PostgreSQL 18 backup operations.

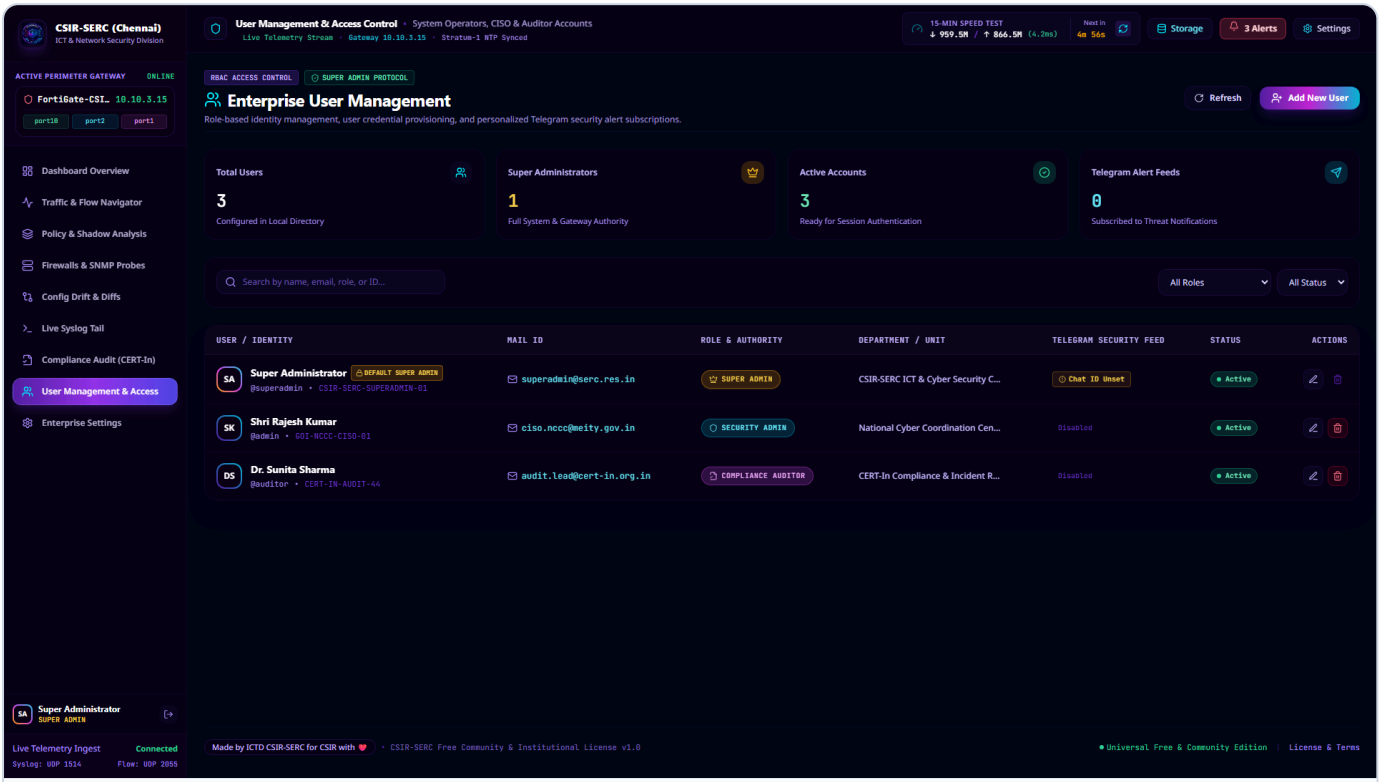


Figure 9: User Management & Telegram Alert Configuration

RBAC

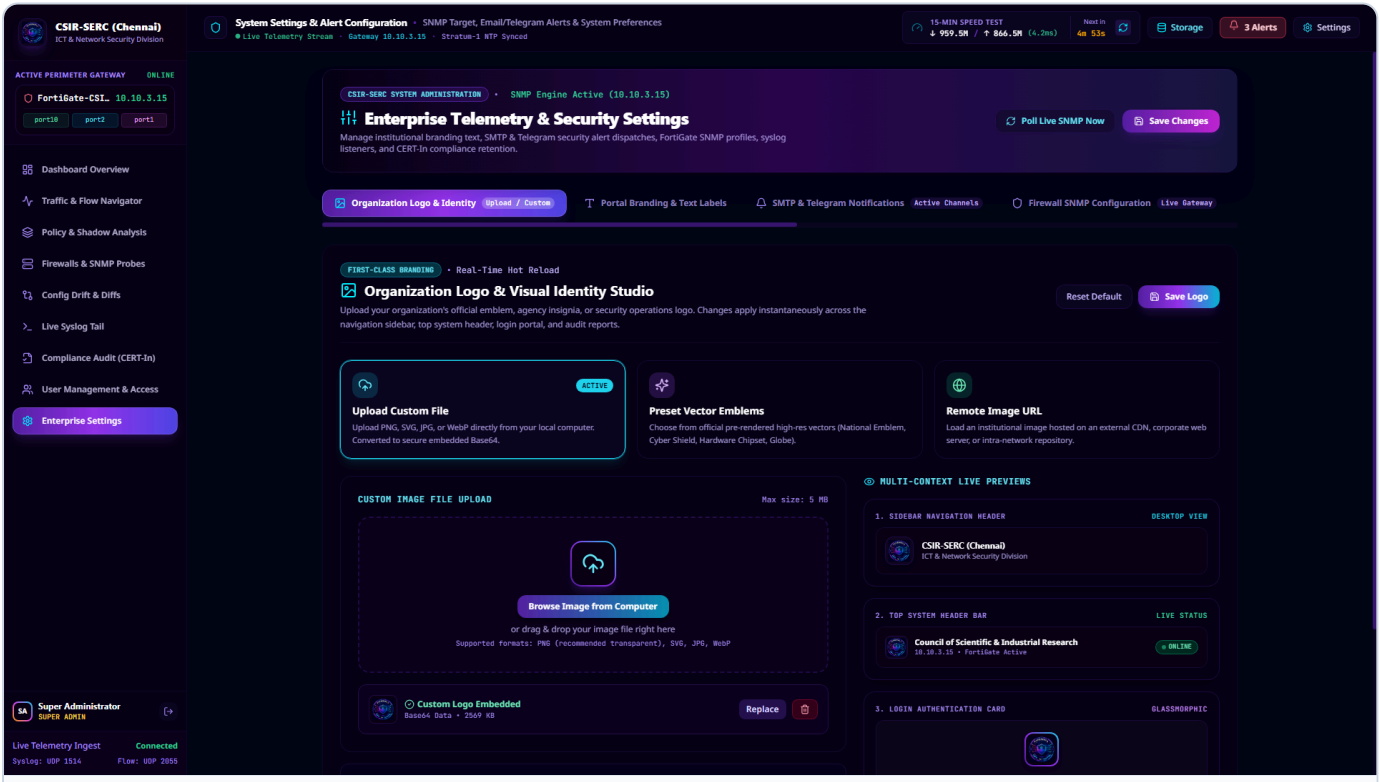


Figure 10: Institutional Branding, Custom Logo & PostgreSQL 18 Backup Scheduler

ADMINISTRATION

3. Technical Specifications & Reference Tables

3.1 FortiGate Hardware Sensor OID Reference

Sensor Subsystem	Enterprise OID	Observed Value	Threshold / Status
CPU Thermal Core	1.3.6.1.4.1.12356.101.4.3.2.1.3 (DTS0)	46°C	Normal (< 75°C)
Board Center Thermal	1.3.6.1.4.1.12356.101.4.3.2.1.3 (TD1)	36°C	Normal (< 65°C)
Chassis Exhaust Thermal	1.3.6.1.4.1.12356.101.4.3.2.1.3 (TD2)	44°C	Normal (< 70°C)
NP6 Accelerator Thermal	1.3.6.1.4.1.12356.101.4.3.2.1.3 (TR3)	40°C	Normal (< 70°C)
Cooling Fan 1	1.3.6.1.4.1.12356.101.4.3.1.1.3 (Fan 1)	6,400 RPM	Normal (Active)
Cooling Fan 2	1.3.6.1.4.1.12356.101.4.3.1.1.3 (Fan 2)	6,400 RPM	Normal (Active)
Cooling Fan 4	1.3.6.1.4.1.12356.101.4.3.1.1.3 (Fan 4)	5,200 RPM	Normal (Active)
Power Supply 1 (Primary)	1.3.6.1.4.1.12356.101.4.3.3.1 (PSU 1)	228V AC, 48W	Normal (Dual Redundant)
Power Supply 2 (Redundant)	1.3.6.1.4.1.12356.101.4.3.3.1 (PSU 2)	230V AC, 16W	Normal (Dual Redundant)

3.2 Top FortiOS Security Policy Accounting

Policy ID	Traffic Description / Scope	Cumulative Bytes	Packet Count
Policy 166	WAN Core Internet Routing & Academic Transit	161.58 TB	172.8 Billion
Policy 76	Datacenter Core Inter-VLAN & High-Speed Switch	36.60 TB	37.3 Billion
Policy 1	Default Perimeter Outbound NAT	11.45 TB	17.8 Billion
Policy 156	Campus Research Network & Supercomputer Egress	5.98 TB	6.2 Billion
Policy 84	High-Performance Computing Cluster (HPC) Tier	5.95 TB	6.36 Billion

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